



Krishna Kankipati

AI architect and engineer building resilient, interpretable systems that make advanced AI useful in real-world work.

Dallas-Fort Worth, Texas

[LinkedIn](#) [GitHub](#) [Portfolio](#)

Practice

I work at the intersection of large-scale systems engineering and AI research. My goal is to make advanced AI accessible and useful by building products that remain resilient, interpretable, and trustworthy in real-world use.

As a founding AI engineer at SpendRule, I turn complex healthcare agreements into traceable rules for invoice validation. My broader work spans legal AI, healthcare, finance, public sector search, and computer vision. I design the full path from messy source data to a dependable product: tenant-aware intake, grounded retrieval and model outputs, deterministic safeguards, event-driven execution, traces, measurable quality, and workflows that recover safely when something fails.

Hackathon awards

SafeScreen AI 1st place [Team entry ↗](#)
Qualcomm × Meta ExecuTorch Hackathon · Jun 2026

Built with a team: an Android visual safety layer that detects explicit content locally and blurs it across apps. ExecuTorch runs the on-device models; the app does not request internet permission. [See the PyTorch winners recap ↗](#)

ARIA 2nd place · prize track [Team entry ↗](#)
AI Trading Agents Hackathon · Apr 2026

Built with team 0TruXt: a multi-strategy trading agent using Kraken paper trading, 13 code-enforced risk gates, and ERC-8004 records for verifiable decisions. [Explore the code ↗](#)

Experience

Founding AI Engineer Dec 2025 – present
SpendRule

- Architected a multi-tenant contract and invoice platform that binds financial decisions to governing source passages, effective dates, facilities, and services.
- Built SFTP and S3 dropzone intake with archive and file event handlers, deduplication, version-aware document handling, and a content-addressed vault.
- Designed durable, event-driven extraction, compilation, validation, audit, and recovery stages with checkpoints, retries, and explicit processing state.
- Separated AI-based semantic matching from deterministic pricing and policy rules; ambiguous cases route to review rather than receiving an unsupported verdict.
- Added tracing, quality evaluation, asynchronous audit, and guarded dry runs to make the system inspectable and safer to operate.

AI Engineer

Jan 2024 – Jan 2026

108-AI / Casey Legal

- Built legal AI services spanning document ingestion, OCR, chunking, hybrid retrieval, and grounded single- and multi-document Q&A.
- Developed stateful contract review and playbook-customization workflows that fetch tenant context, retrieve relevant clauses, review provisions, draft alternatives, validate outputs, and persist results.
- Wrapped long-running AI work in event-driven execution with retries, status and failure handling; instrumented graph nodes and model calls with tracing for diagnosis and evaluation.
- Worked on reusable agent interfaces for comparison, redaction, summarization, compliance review, and document exports across containerized Azure services.

AI Developer

Sep – Dec 2023

OpenView Partners

- Built predictive analytics and forecasting pipelines in Snowflake using Snowpark and Ray for investment research.
- Developed a BERT based recommendation engine and natural language to SQL tools for exploring company and warehouse data.
- Deployed modular LLM applications on GCP with FastAPI, Cloud Run, Pub/Sub, and Cloud Functions.

AI & Web Engineer Intern

Sep – Dec 2022

NolymitAI

- Built a multilingual restaurant chatbot with Dialogflow, translation, speech, and OCR services.
- Deployed a TypeScript and Node.js web interface with Firebase authentication; built additional small-business features including content detection.
- Coordinated weekly plans and progress reporting across the project team.

Graduate Course Assistant

Sep – Dec 2022

Stevens Institute of Technology

- Supported more than 50 students in a graduate algorithms course through instruction and individual guidance.
- Helped develop and evaluate assignments and examination materials.

Computer Vision Engineer

May 2018 – Aug 2021

Vayu Drone Technologies

- Built geospatial analysis workflows for drone-based land surveys, terrain classification, and environmental monitoring.
- Trained and deployed image segmentation and object detection models using PyTorch, TensorFlow, and OpenCV.
- Developed data services with PostgreSQL/PostGIS and AWS for ingestion and analysis of aerial imagery; processed more than 500 images daily.

Consulting work

Investment analysis platform

Finance

ExMachina Capital

- Architected and built a Spring Boot and AngularJS application for investment and stock analysis, organizational research, deal rooms, and M&A workflows.
 - Added an AI assistant for analysts and helped move the platform from concept to application over eight months.
-

Intelligent policy search

Public sector

ALT Solutions · Housing & Urban Development

- Architected search across more than 2,000 mixed format policy documents using AWS Bedrock, OpenSearch, SageMaker, LangChain, Unstructured, and PyMuPDF.
- Enabled semantic navigation of unstructured documents, improving analyst efficiency by 50%.

Healthcare support copilots

Healthcare

ARS

- Delivered customized language model APIs for support, appointment scheduling, and recovery center matching.
- Used SageMaker, Lambda, and API Gateway, with a focus on reliability and handling sensitive information.

System index

The homepage examines four flagship systems in detail. This index includes those systems and the broader project inventory; status describes the work, not a claim of commercial deployment.

Flagship systems

SpendRule

Contract intelligence /
healthcare

SFTP and dropzone events feed a versioned vault, durable AI validation, source-linked decisions, traces, and audit.

Production work

Casey / 108-AI

Legal AI platform

Ingestion and hybrid retrieval support traced review, drafting, and validation graphs inside event-driven workflows.

Professional work

SafeScreen AI

On-device visual safety

Local inference and an explicit intervention policy keep screen frames on-device; uncertain deepfake results stay qualified.

Hackathon / 1st place

ARIA

Autonomous agent / finance

Code-enforced risk gates separate agent proposals from paper-trade permission; signed records explain accepted and rejected actions.

Hackathon / 2nd in a
prize track

Research and product explorations

Chronoscope

Historical visual search

Exact image crops and OCR chunks become the evaluation unit; human labels anchor search-quality claims.

Active exploration

MITRA

Indian Constitution search

Hierarchical, citable source material supports navigation and grounded constitutional Q&A.

Active project

DevCrush

Terminal-native social product

Playful developer matching with explicit house-bot labels, opt-in profile data, and a privacy-conscious local scanner.

Active product

Document analysis

Multimodal retrieval

Visual embeddings and document Q&A explored with grounded evaluation and offline options for sensitive use.

Active exploration

Interfaces and infrastructure

Excalidraw G.R.I.D.

Visual workspace

Diagram organization and AI-assisted edits in an Excalidraw fork.

Public repository

ColNomic embedding API

Retrieval infrastructure

Text and image embedding service with offline loading, adapter support, validation, and health checks.

[Public repository](#)

BugFlow

Security workflow

Report triage with AI classification, semantic duplicate detection, access roles, and background processing.

[Public repository](#)

Giffify

Agent tooling / media automation

A portable Codex and Claude Code skill that turns timestamped video clips into optimized GIFs. A deterministic FFmpeg engine handles validation, batch manifests, output planning, and structured results.

[Public repository](#)

Browse public repositories

Technical skills

LLM systems

Structured tool outputs, evidence-grounded generation, abstention and review paths, multi-provider integration, RAG, AI agents, MCP, LangGraph, LangChain, LlamaIndex, PaperQA2, evaluation and cost tracing

Model adaptation

SFT, PEFT, RLHF, DPO, PPO and GRPO; Hugging Face, PyTorch, TensorFlow, scikit-learn, experiment tracking with MLflow and W&B

Computer vision

Object detection, segmentation, image classification, multimodal embeddings; YOLO, YOLACT++, Faster R-CNN, UNet, OpenCV, ONNX and ExecuTorch

Data & security

Multi-tenant isolation, versioned records, source provenance, guarded data changes, PostgreSQL/PostGIS, vector search, Snowflake and PySpark

Distributed systems

Durable event-driven execution, idempotency, retries, checkpoints, failure recovery, concurrency control, FastAPI, Inngest and Prefect

Cloud & delivery

AWS (SageMaker, Bedrock, Lambda, S3, OpenSearch), Azure, GCP (Vertex AI, Cloud Run, BigQuery), Docker, Terraform, CI/CD, Langfuse and LangSmith

Application code

Python, TypeScript, React, Next.js, Spring Boot, API design, testing and observability

Education

M.S. in Machine Learning

Stevens Institute of Technology · Hoboken, New Jersey

Aug 2021 – May 2023

B.Tech. in Computer Science

VRSEC · Andhra Pradesh, India

Credentials & writing

- **Vision Language Models Bootcamp** · OpenCV University, 2025
- **Generative AI with Large Language Models** · DeepLearning.AI, 2023
- **LangChain & Vector Databases in Production** · Activeloop, 2023
- **Computer Vision Nanodegree** · Udacity, 2020

Published an introduction to topological superconductivity and Majorana fermions for quantum engineers, and articles on practical machine learning. [See writing on LinkedIn ↗](#)